NETWORKING

Bridging On-Prem and Cloud

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Agenda

01 **Why cloud**

tens of gigawatts this decade — multiple providers added since 2025

02 **The problem: heterogeneity**

same topology name, a different implementation at every cloud

03 **Solutions to shrink onboarding time and improve performance**

1	2	3	4
Share Meta's AI learnings	**Standardized benchmarks**	**Workload-representative**	**Unified telemetry**
at design / due-diligence	early, at the vendor	at turn-up, then tune the fabric	across on-prem and cloud

04 **Results**

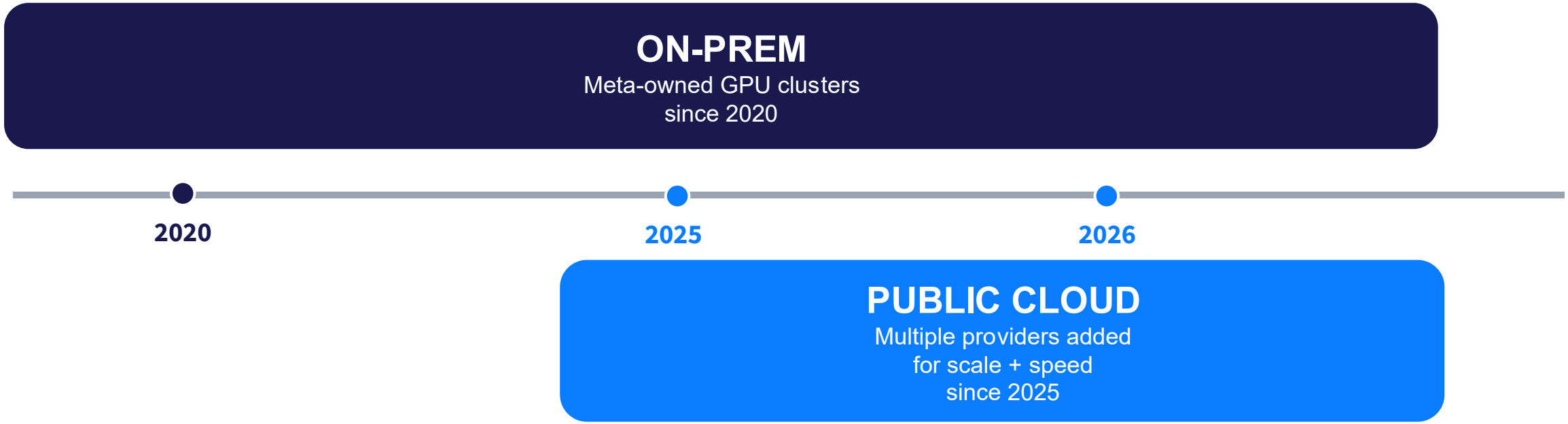
5× faster to onboard a cloud than our first

05 **Call to action**

make hybrid AI networking work the same — everywhere

Why hybrid?

Meta has already announced tens of gigawatts this decade — Prometheus ~1 GW (2026) · Hyperion scaling to 5 GW



What's different about Cloud infrastructure?

Cloud networking is fundamentally different from owning the rack — new constraints, topologies, & operational challenges

Different GPU Vendors / SKUs

Different Network Topologies

ToR / Multi-plane

Different Fabric Technologies/Transports

InfiniBand / RoCE / EFA

Example: Heterogeneity in Cloud Transports

CLOUD	TOPOLOGY	LOAD BALANCING	CONGESTION CONTROL	TRANSPORT
CLOUD A	RAIL-OPTIMIZED	ECMP HASHING	CUSTOM PROGRAMMABLE CONGESTION CONTROL	LOSSY TRANSPORT WITH AGGRESSIVE RETRANS
CLOUD B	TOP OF THE RACK	FLOWLET LOAD BALANCING	DCQCN	PFC / LOSSLESS
CLOUD C	MULTI-PLANAR	PACKET SPRAYING	SPECTRUM-X CC	PFC / LOSSLESS

Our Solution to Scale Cloud Life Cycle

ALIGNING FABRIC DESIGN / CONFIG BY SHARING META'S AI LEARNINGS WITH CLOUDS

Design

Share our experience operating RDMA clusters as part of design due diligence

Customize CSP's RDMA Network configuration as necessary

RUN STANDARDIZED BENCHMARKS EARLY, AT THE CLOUD VENDOR

Bring-up

Vendor runs Collective sweeps across GPU gang sizes / message sizes.

Meta introduced Collective Performance Guarantees into Contract language

RUN WORKLOAD REPRESENTATIVE PRIMITIVES ON TURNUP

Handoff

Characterizes workload representative patterns

Run them prior to handing them off to Internal PGs

Co-design solution with CSP

BRIDGE ON-PREM VS CLOUD TELEMETRY

Sustaining

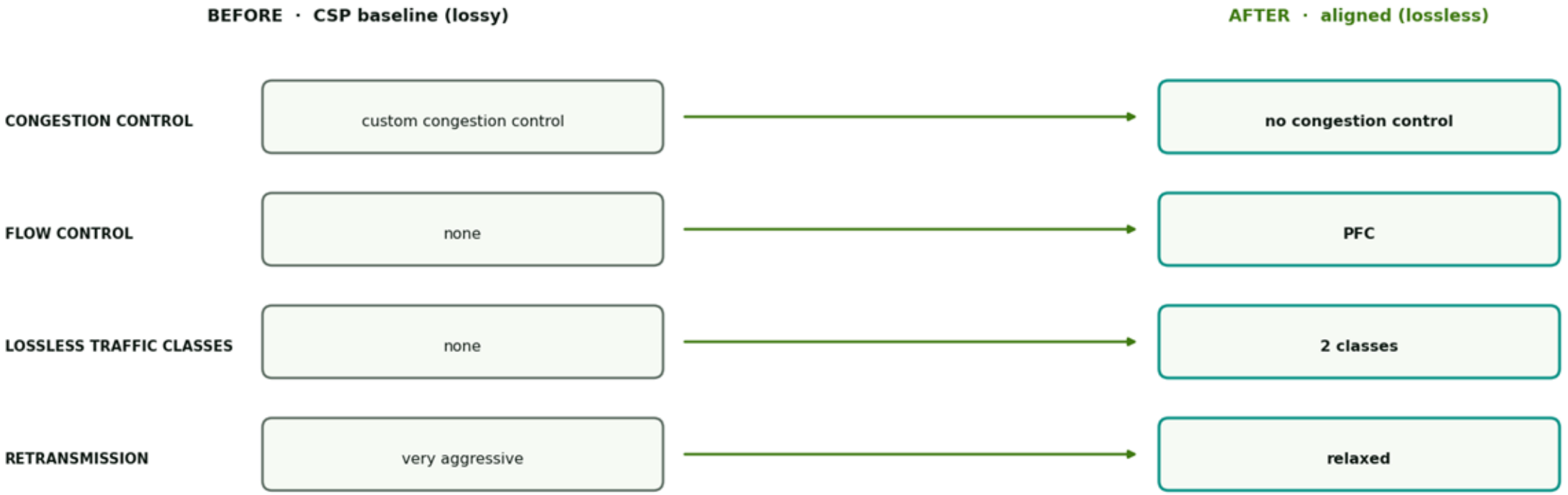
Compensate for missing switch access / telemetry

Run Passive monitoring on Clouds.

Bespoke API to consume switch counters

1 · Cloud Due Diligence: Aligning Transport for On-Prem Parity

The CSP's lossy transport was proven at scale for their own workloads — a different design point.
Meta's internal customers needed this capacity to behave like an on-prem cluster.



What moved the decision

six years of Meta experience configuring and operating lossless fabrics, shared openly

Outcome

parity with on-prem clusters

2 · Run Standardized Benchmarks Early, At the Vendor

Standardized Performance Validation Framework

Partnered with cloud vendors to execute comprehensive point-to-point and collective testing prior to cluster handoff.

✓ **Enforces Transport & Collective performance expectations in Contract Language**

01

Point-to-Point Latency & BW

P2P Microbenchmarks: Bandwidth and latency tests using ib_write_bw and ib_write_lat over IPv6.

Multi-Hop Traversal: Test across Leaf, Spine, and Super-Spine network layers to measure hop-by-hop degradation.

02

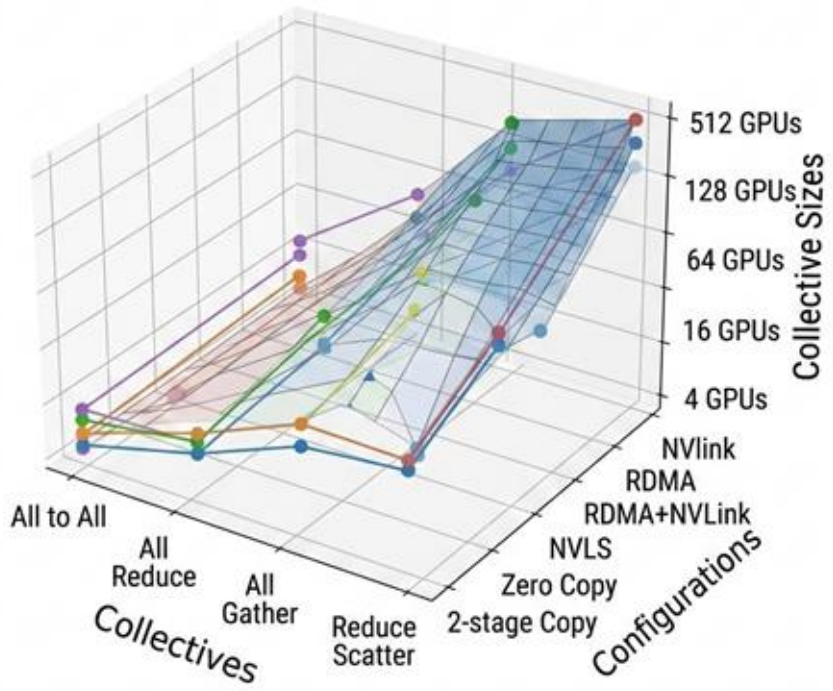
Collective Benchmarking at Scale

NCCL / RCCL Collectives: Measure key operations including Alltoall, Allreduce, Allgather, and ReduceScatter.

Hierarchical Scaling: Execute test profiles scaling dynamically from 16 to 512+ GPUs under precise CCL configurations.

Additional long duration collective tests to measure stability

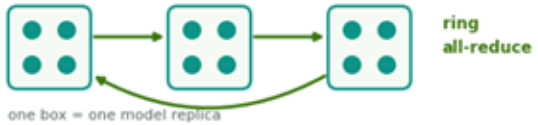
Collective Performance Sweeps at the Vendor



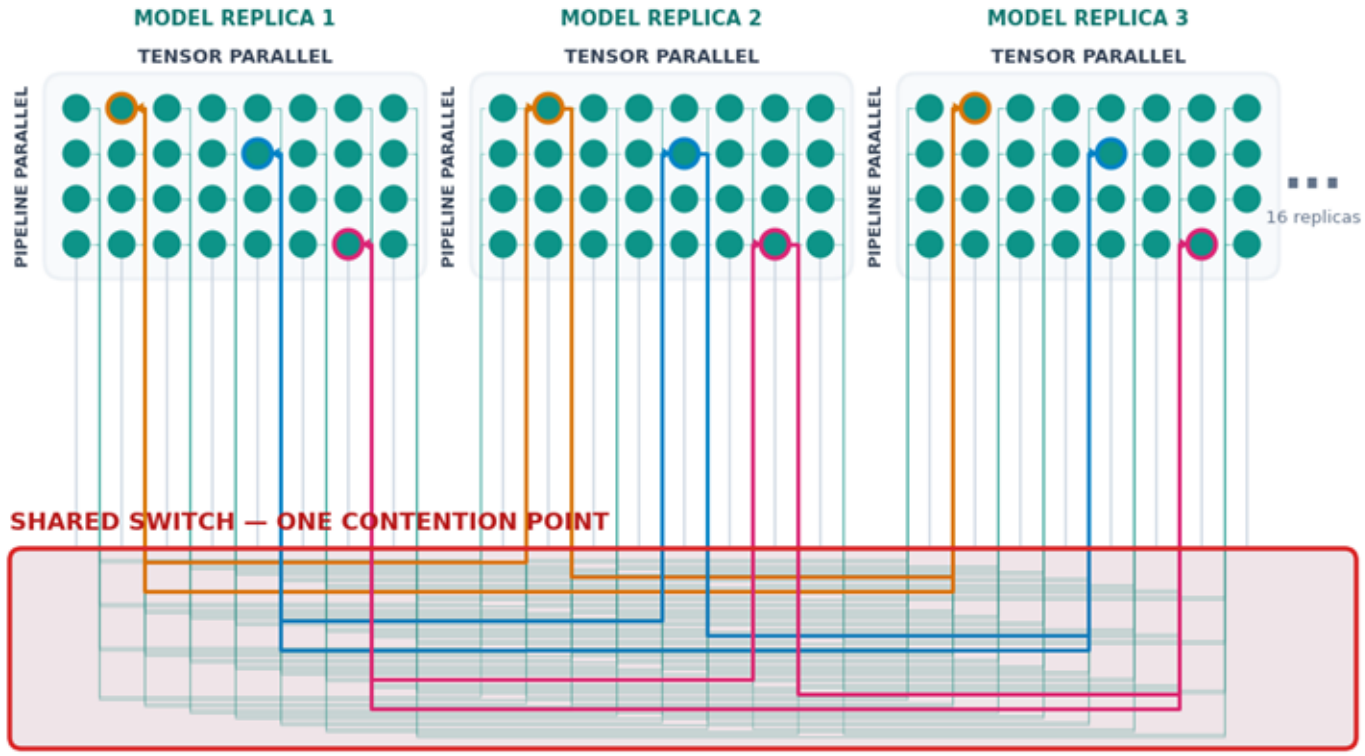
3 · Evolution of RDMA Traffic & Software Stacks

	PAST YEAR	EVOLVED TO PRESENT
WORKLOADS	**Pre-training Dominating** Single family of workload type Operating over dedicated network resources. 2 use-cases: Ranking/Recommendation and GenAI Pre-training	**3X Workload Diversity** Post-training, Inference scaling rapidly along with Pre-training
COMMUNICATION SOFTWARE	**1–2 Standardized Libraries** Simple software ecosystem directly accessing RDMA verbs in production.	**6x RDMA Verbs Consumers** Diverse software stacks, custom transport layers, and specialized collective frameworks.

3 · Run Evolving Workload Traffic Graphs on Internal Cloud turn-up

WORKLOAD	TRAFFIC PATTERN	NETWORK CONGESTION	TUNING NEED
PRE-TRAINING	**CROSS-REPLICA ALL-REDUCE** 	**MANY SIMULTANEOUS COLLECTIVES** on the same network switch	**TUNE LOAD BALANCING**
POST-TRAINING	**WEIGHT TRANSFER, TRAINER → SAMPLER** 	**INCAST** many senders, one receiver	**PFC BUFFER CONFIGS** reduce over-subscribed headroom
INFERENCE	**REMOTE KV CACHE WRITTEN VIA CPU TO RDMA NIC** 	**SLOW RECEIVER** RDMA NIC exerts PFC upstream	**TUNE CONGESTION CONTROL** + comms library configuration to reduce rate

3 · Tuning Multiple Replica Pre-training Workload after Cloud turn-up



All Reduce (Data Parallel) Synchronization

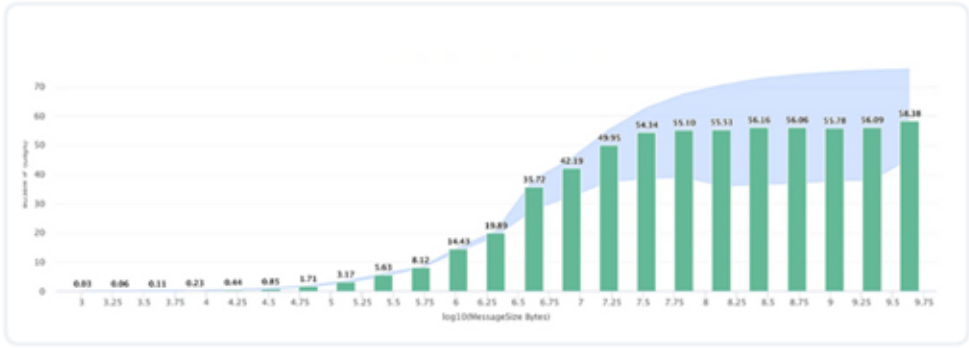
Tuned flowlet + CC: variability collapses and peak bandwidth climbs 29%

each ring spans 16 GPUs — one per replica · 32 rings per replica set

MEASURED ON CSP FABRIC

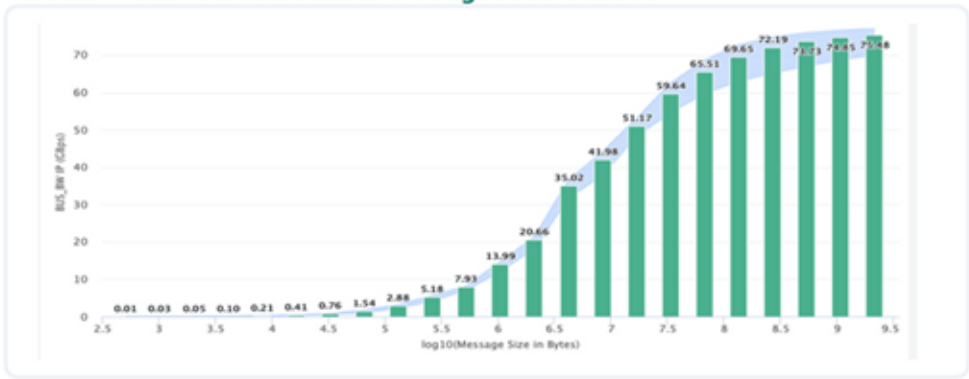
32 parallel rings · 16 GPUs per ring · NCCL bus BW vs message size

BEFORE — default flowlet timer + congestion control



wide shaded band = large run-to-run variability across the 32 rings

AFTER — tuned flowlet timer + congestion control



band collapses — runs land on top of each other

Peak bus BW 58.4 → 75.5 GB/s

+29% at large messages, with variability largely gone

4 · Unifying On-Prem and Cloud Telemetry

WHAT WE HAVE

Full host access

Baremetal access in every cloud. We get endpoint counters, same as on-prem.

OBSERVABILITY CHALLENGES

The switches are dark to us

Switch telemetry only reaches us if the provider agrees to send it. Every CSP needs a contract covering the metric set, the format, and the frequency.

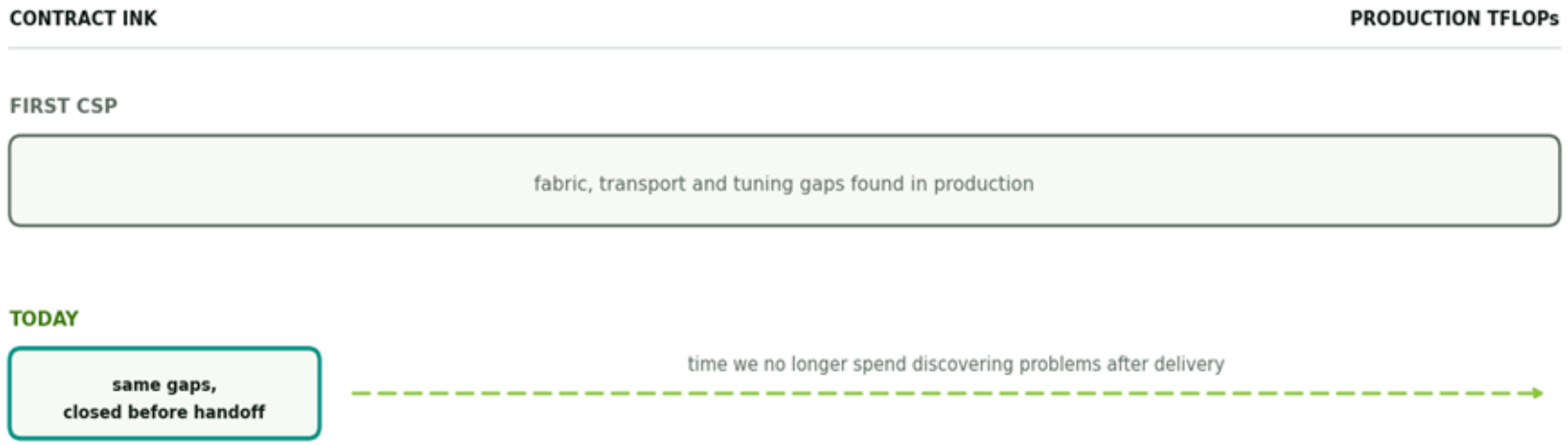
Formats differ per provider, so debugging is bespoke. A pinger gives us some black-box coverage, but every fabric issue still takes complex coordination with the CSP.

Putting It Together: Design to Handoff

5×

FASTER

to onboard a cloud today
than our first CSP onboarding



WHAT COMPRESSED IT

1 Learnings shared at design

config delta resolved before hardware lands

2 Benchmarks at the vendor

collective guarantees in contract language

3 Tuning at turn-up

workload-representative patterns, not synthetic traffic

4 Telemetry bridged

passive monitoring where switches are dark

Measured earlier in this talk: tuned flowlet timer + congestion control — 58.4 → 75.5 GB/s, run-to-run variability collapsed

Call to Action

Make hybrid AI networking work the same — everywhere

Standardize RDMA Testing

Standardized set of tests, that CSPs can run prior to delivering the cluster

Standardize Workload Testing

Get a representative repository of traces for various kinds of workloads (pre-training, post-training inference)

Shared telemetry

Uniform observability wherever the GPUs live

Build it with OCP: Partner with OCP workgroups to build open standards for testing and telemetry

Thank You!

